Auditing Bias in AI Resume Screening

Does removing demographic information make LLM hiring decisions fairer?

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ECE 4424 / CS 4824

WHAT WE KNOW SO FAR

A matched-pair screening audit is implemented.

An actual LLM fairness conclusion is not yet supported.

THE EVIDENCE

11 matched pairs

22 synthetic resumes

8 held-out evaluation pairs

OFFLINE SANITY TEST: 0 score gap; 48/48 skill-stress checks pass

Zero gap is expected because the offline heuristic ignores names.

These results validate code behavior, not the fairness of an LLM.

WHY IT MATTERS

Recruiters use resume screening to decide who receives further human review. Job-related qualifications should matter more than demographic cues.

HOW WE TESTED IT

Baseline: identical screening prompt with named resume. Intervention: candidate name replaced by a neutral placeholder. Measures: paired score gaps, Yes/No differences, and qualification sensitivity.

NEXT TEST + CLASS CONNECTION

Run repeated actual LLM calls on held-out cases; compare indirect cues and weaker qualifications. Class theme: trust requires testing under changing conditions.

WHERE THE CLAIM STOPS

Small synthetic dataset, name as an imperfect proxy, no human ranking labels, and no completed LLM outcomes. Do not use to automate hiring.

Sources: EEOC (2024); Tripathi et al. (2026); Gerszberg et al. (2026). Code and report: see team project package.